

ZEERBOX · DESIGN DECISION RECORD

ZeerBox TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/zeerbox		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 8); items 9 and 10 and the new TRL 3 items 11 to 13 remain proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction approved cross-cutting SwapCell interface items, a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 section) and ZBX-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Budget	Decided by Amish, 2026-09-25: go with recommendation. Option (a): keep <code>budget_usd: 300</code> and redefine it as the cooling equipment kit (items 6 to 11 and 14). The structure is built from local materials and costed separately. R12 is redefined to match	<code>project.yaml</code> (budget unchanged), ZBX-REQ-001 v0.3 R12, <code>bom/bom-notes.md</code>
2	Solar panel size	Decided by Amish, 2026-09-25: go with recommendation. 150 W panel	<code>bom/bom.csv</code> line 10, <code>cad/src/model.py</code> , ZBX-CAL-001 section 7
3	Battery	Decided by Amish, 2026-09-25: go with recommendation. Keep the 12.8 V 12 Ah LiFePO4 battery for evening ventilation	<code>bom/bom.csv</code> line 11
4	Forced air through a pad	Decided by Amish, 2026-09-25: go with recommendation. Forced air, not a passive ZECC-style chamber	ZBX-PRC-001 v0.3
5	Fan arrangement	Decided by Amish, 2026-09-25: go with recommendation. Exhaust fans beside the door, pad on the opposite wall	ZBX-PRC-001 v0.3, <code>cad/src/model.py</code>
6	Wall type	Decided by Amish, 2026-09-25: go with recommendation. Wet sand cavity walls, pending the TRL 3 heat balance. That heat balance (ZBX-CAL-001 section 4.1) shows only a small benefit; see open item 12	ZBX-PRC-001 v0.3, <code>cad/src/model.py</code>
7	Store size	Decided by Amish, 2026-09-25: go with recommendation. Keep 24 crates until a partner confirms the need	ZBX-REQ-001 R1, <code>cad/src/model.py</code>
8	Control thresholds	Decided by Amish, 2026-09-25: go with recommendation. 20 °C set point, 4 K wet-bulb depression and 95 % RH guard as starting values for field tuning	ZBX-PRC-001 v0.3, control modes

The pitch and problem lines had no recommended rewording in the TRL 2 review and are unchanged.

Cross-cutting approvals from the same instruction, recorded here:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell interface adds a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles. ZeerBox does not use SwapCell (its 31 W, 12 V load runs from a 154 Wh battery), so no v0.3 items apply.
- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting). Not applicable: ZeerBox has no SwapCell pack.
- **Co-design partners per area later.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. The partner for ZeerBox stays open (item 10).

Items that remain open

Table 2. Open items, proposed, awaiting Amish.

#	Item	Why it is open	Options and recommendation
9	Wall material	The TRL 2 review listed fired brick against mud brick or block with no recommendation	Fired brick (the working choice in the model and BOM), mud brick or concrete block. No recommendation stated
10	First partner, region and crop mix	No recommendation; the cross-cutting rule leaves partners to be picked per area later	For example a dry-season vegetable area in the Sahel, or a group already using D-Lab designs in Kenya
11	R3 not met, R2 at risk	New at TRL 3: store RH is about 71 % against 85 %, and the unfavorable case gives 8.0 K against 8 K (ZBX-CAL-001 sections 4 and 5)	(a) 150 mm pad on the same fans: 10.5 K and 81 % RH, about \$10 more (indicative); (b) relax R3 to 70 % and add produce-level measures such as lined or covered crates; (c) both. Recommendation: (c), a 150 mm pad and R3 at 80 %
12	Wet cavity against a dry insulating fill	New at TRL 3: the wet cavity gives 0.15 K over dry sand for about 12 L of water a day, and is slightly worse than dry sand in the unfavorable case; dry rice husk gives 9.2 K with no water (ZBX-CAL-001 section 4.1)	(a) keep the wet cavity as decided; (b) dry rice husk or similar fill, kept dry and protected from termites and rodents; (c) dry sand. Recommendation: (b), subject to a durable, pest-proof fill being available locally
13	Roof footing size and design wind speed	New at TRL 3: at a 30 m/s gust each post needs about 2.2 kN of hold-down; the model uses 500 x 500 x 600 mm footings (factor 1.6)	Confirm the local design wind speed with a builder. Recommendation: keep 500 mm footings as a minimum

■ Consequences

- R12 now covers the \$300 cooling equipment kit; the kit costs \$260 and the structure, about \$360, is costed separately.
- The 150 W panel meets R7 with a 41 % margin, but it needs a 20 A charge controller and 2.5 mm² PV cable (ZBX-CAL-001 section 7). Both are in the BOM.
- R3 is not met and R2 is at risk until item 11 is decided.
- TRL 4 work (bench tests of the controller, a pad and fan test article, build procedures) is on hold by Amish's instruction.